Brazing Torch



Category: **Welding & Soldering** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Brazing tool; Brazing torch

What it is used for

A brazing torch heats compatible metal parts and filler material so that the filler flows into a joint without melting the base metals. Correct flame adjustment, joint preparation, and ventilation help create reliable joints while reducing fire and burn risks.

Main parts

- Torch body and handle provide controlled gripping and operation.
- Fuel gas connection supplies the selected fuel gas.
- Oxygen or air inlet supports combustion depending on the torch design.
- Hoses carry gases from compatible supply equipment.
- Regulators control gas pressure where fitted.
- Torch tip or nozzle shapes the flame.
- Control valves regulate gas flow to the torch.
- Ignition device or approved striker lights the torch.

Before you start

1. Only trained personnel should operate a brazing torch under the approved workshop procedure.
2. Read the torch and gas supply instructions and confirm fuel, tip, and regulator compatibility.
3. Inspect hoses, connections, regulators, valves, torch body, and nozzle for damage or leaks.
4. Secure gas cylinders upright and away from heat, sparks, and traffic.
5. Clean and fit the joint parts, and select compatible filler metal and flux where required.
6. Remove combustible materials, provide ventilation, and prepare suitable fire protection and PPE.

How to use

1. Check that gas valves are closed before connecting the torch to the approved supply.
2. Connect the correct hoses and regulators and secure all fittings as instructed.
3. Check connections for leaks using an approved leak-detection method before ignition.
4. Set the gas supply and regulator settings according to the torch instructions.
5. Ignite the torch using the approved ignition device and adjust the flame as specified.
6. Heat the joint area evenly and avoid directing the flame toward hoses or cylinders.
7. Apply compatible filler metal and flux as required, allowing the filler to flow into the joint.
8. Stop heating when the joint is filled and avoid overheating the base metal.
9. Close the torch valves and gas supply in the sequence specified by the manufacturer.
10. Allow the joint and torch to cool safely, then inspect the joint and store equipment properly.

⚠ Safety

- Wear suitable eye protection, heat-resistant gloves, protective clothing, and safety footwear.
- Never use a torch with damaged hoses, leaking connections, or faulty valves.
- Keep cylinders upright, secured, and away from hot work and ignition sources.
- Never use oil or grease on oxygen fittings and keep gas equipment clean.
- Provide adequate ventilation and avoid inhaling flux fumes or heated-metal fumes.
- Keep a suitable fire extinguisher available and check the work area for hot spots after brazing.

Care & storage

- Close gas supplies and allow the torch to cool before cleaning or storage.
- Inspect hoses, regulators, valves, and the nozzle regularly for wear or blockage.
- Clean the tip using the recommended tools and avoid enlarging the nozzle opening.
- Store the torch and accessories in a dry, secure area and arrange qualified servicing for faults.

Common mistakes to avoid

- Using incompatible fuel, oxygen, regulator, or torch tip combinations.
- Lighting the torch before checking connections for leaks.
- Overheating the base metal instead of allowing filler metal to flow properly.
- Leaving the gas supply open or the work area unchecked after brazing.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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